

Diligent Code Studio

Community Edition User Manual

Version 0.5.5 - Help Button and Built-in PDF Manual

Purpose	A local-first development workbench for editing code, managing projects, using AI assistance, checking dependencies, running builds, and preparing releases.
Audience	Developers, IT administrators, power users, and Diligent Software Services project maintainers.
Platform	Windows-focused desktop workflow with cross-platform source support through Tauri, React, TypeScript, Rust, and Vite.

This manual explains how to navigate the application, use the Workspace Menu, configure setup dependencies, work with the editor, access AI help, use Git, run diagnostics, and package releases.

Diligent Software Services

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1. Overview

What Diligent Code Studio is

Diligent Code Studio is a desktop software-building workbench designed to keep the common development tasks in one place. It combines project navigation, editing, AI assistance, dependency checks, Git tools, diagnostics, release packaging, and logs.

The application is designed around a local-first workflow. Your project files remain on your computer. AI features are optional and must be configured in Settings before use.

Primary goals

- Make project setup easier by checking required tools and dependencies.
- Give the user a clean editor with project navigation and file actions.
- Provide compact AI help on every screen without taking over the workspace.
- Help users run terminal commands, diagnose errors, manage Git, and package releases.
- Keep the interface beginner-friendly while preserving advanced project controls.

2. First Launch

Recommended first steps

- Open the application from the installed shortcut or compiled executable.
- Choose a workspace folder from the left Workspace panel.
- Open Setup & Dependencies and select Check Again to verify tools.
- Open Settings to choose theme, compact mode, terminal shell, and AI provider.
- Use the Help button in the top-right corner to open this PDF manual inside the software.

Important note

The development command `npm run tauri:dev` is only for testing from source. The installed desktop version should run as a normal application and should not require an npm window to remain open.

3. Main Layout

Application zones

- Top app bar - shows the application name, project health indicators, Setup, Help, and AI Help controls.
- Left sidebar - workspace path, file actions, filter box, and project file tree.
- Top Workspace Menu - main navigation buttons for Templates, Setup, Editor, AI, Search, Terminal, Git, Problems, Release, Tool Registry, Project Dashboard, Logs, and Settings.
- Main content area - shows the selected screen.
- Compact AI Help - floating assistant available on every screen.

Why the terminal is not global

The terminal is intentionally no longer open on every screen. It is available from the Terminal page so normal screens stay clean and focused.

4. Help Button and PDF Manual

Opening the manual

Version 0.5.5 adds a Help button in the top-right app bar. Selecting Help opens this PDF manual inside a built-in help viewer overlay.

Help viewer controls

- Open PDF - opens the manual PDF in a new viewer window or browser-capable view if supported by the system WebView.
- Close - returns to the current Diligent Code Studio screen.
- Scroll - use the PDF viewer scroll controls to move through this manual.

Where the PDF is stored

The manual is included with the application files under public/manuals in the source package and is copied into the built frontend output during the normal Vite build process.

5. Workspace Menu

Purpose

The Workspace Menu is the main navigation area. It stays in its own fixed top space and does not horizontally scroll. Buttons wrap naturally when needed.

Drag-and-drop ordering

- Drag a Workspace Menu button left or right to reorder it.
- Drop on the left half of another button to place it before that button.
- Drop on the right half to place it after that button.
- The new order is saved to preferences automatically.
- Use Reset Menu Order in Settings to restore the default order.

Default order

Templates is designed to start farthest left by default, followed by Setup & Dependencies, Editor, AI, Find/Search, Terminal, Git, Problems, Release, Tool Registry, Project Dashboard, Logs, and Settings.

6. Workspace and File Explorer

Choosing a workspace

Use the Choose Folder button in the left sidebar to select the project folder you want to work on. Diligent Code Studio scans the folder and displays files and folders in the tree.

File actions

- New File creates a file in the selected folder or current target directory.
- New Folder creates a folder in the selected target directory.
- Rename renames the currently selected file or folder.
- Delete removes the selected file or folder after confirmation.
- Refresh reloads the workspace tree.

Filtering files

Use the filter box above the file tree to quickly find files or folders by name.

7. Editor

Opening and editing files

Select a file in the left file tree to open it in the editor. Open files appear as tabs. Unsaved changes are marked and can be saved from the toolbar.

Editor toolbar

- Save - saves the active file.
- Save As - writes the active file to a new location.
- Format - formats supported document types when available.
- SHA - creates a SHA-256 hash for the active file.
- AI Help - opens or minimizes compact AI help.

Status strip

The footer under the editor shows the active file path, detected language, line count, cursor location, save status, and hash status.

8. Compact AI Help

Purpose

Compact AI Help is available on every screen but no longer takes up a large permanent panel. It can be minimized to a small bottom-right AI Help button or expanded when needed.

AI quick actions

- Explain - prepares a prompt to explain selected code or the active file.
- Bugs - prepares a prompt to review the current file for problems.
- Errors - focuses the AI context on diagnostics and problems.
- Terminal - focuses the AI context on terminal output.
- Git - focuses the AI context on Git status.
- Navigate - asks for help deciding what to do next on the current screen.

Privacy reminder

AI is optional. Configure the provider in Settings. Review any prompt before sending project content to an external provider.

9. Setup & Dependencies

Purpose

Setup & Dependencies checks whether the computer has the tools needed for common software projects.

Typical tools checked

- Git and GitHub CLI for source control and GitHub workflows.
- Node.js and npm for JavaScript, TypeScript, React, and Vite projects.
- Rust and Cargo for Tauri desktop builds.
- Visual Studio Build Tools for Windows native build support.
- WebView2 Runtime for Windows desktop webview support.
- Ollama for optional local AI models.

Installer results

If winget reports that a package is already installed or up to date, Diligent Code Studio treats that as success instead of a failure.

10. Tool Registry

Purpose

Tool Registry is the app catalog for registered tools, commands, templates, scripts, models, and utilities. It is not the Windows Registry.

How to use it

- Review built-in tool definitions.
- Add custom tools or scripts used in your workflow.
- Enable or disable tools depending on the project.
- Keep notes about where each tool is installed or how it is used.

11. Terminal

Purpose

The Terminal page lets you run project commands from the selected workspace. It is intentionally dedicated to one screen so it does not crowd every other screen.

Common commands

```
npm ci
npm run build
npm run tauri:dev
npm run tauri:build
git status
```

External terminal

Use Open Terminal to launch an external terminal window in the current project folder.

12. Git

Purpose

The Git page helps you review repository state, stage changes, commit work, create tags, and understand current branch status.

Recommended GitHub setup

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
gh auth login
gh auth status
```

When GitHub CLI is installed but not detected

Close and reopen Diligent Code Studio so the application receives the updated PATH. If needed, confirm the GitHub CLI path is included in your user PATH.

13. Problems and Diagnostics

Purpose

The Problems page runs diagnostics and collects build or syntax issues into a more readable view. It is useful after TypeScript, Vite, Rust, Cargo, or build errors.

How to use diagnostics

- Open Problems.
- Run diagnostics for the current workspace.
- Review the file, line, column, and message for each problem.
- Use Compact AI Help with the Errors context to ask for a fix plan.

14. Release Builder

Purpose

The Release Builder helps prepare project release packages after the project has been built.

Recommended build sequence

```
npm ci
npm run build
```

```
npm run tauri:build
```

Release outputs

- Compiled application or installer artifacts.
- ZIP package when supported by the build environment.
- SHA-256 checksum file.
- Release notes file.
- Copied bundle files for easier distribution.

15. Settings

Major settings areas

- Appearance - theme and compact mode.
- Editor - font size, word wrap, and save behavior.
- Workspace - startup folder and workspace memory options.
- AI - provider, model, endpoint, context, and confirmation settings.
- Terminal - preferred shell.
- Workspace Menu - reset menu order to defaults.

Resetting the menu

Use Reset Menu Order in Settings to restore the default Workspace Menu order. The reset control is intentionally kept out of the Workspace Menu itself.

16. Recommended Workflow

Daily workflow

- Open Diligent Code Studio.
- Choose or refresh the workspace.
- Open files from the left file tree.
- Edit and save changes.
- Use Problems to diagnose errors.
- Use Compact AI Help to explain errors or review code.
- Use Git to stage and commit changes.
- Use Release Builder when ready to package.

Testing this project from source

```
npm ci  
npm run build  
npm run tauri:dev
```


17. Troubleshooting

GitHub CLI installed but gh is not recognized

Close and reopen PowerShell and Diligent Code Studio. If the command still fails, confirm GitHub CLI is installed under C:\Program Files\GitHub CLI and add that folder to PATH.

npm window opens in installed version

The installed production app should not require an npm window. Confirm that your shortcut points to the compiled Diligent Code Studio executable and not to a development script or npm command.

Workspace menu is not in the desired order

Drag the menu buttons to reorder them. Use Settings, Reset Menu Order to return to the default order.

AI does not answer

Open Settings and configure OpenAI or Ollama. If using Ollama, verify the Ollama app is running and the configured model is available.

18. Keyboard and Usage Tips

Useful habits

- Keep one project per workspace when possible.
- Run Setup & Dependencies after installing new developer tools.
- Use Problems before asking AI about build errors so the assistant has better context.
- Use Git commits after each working milestone.
- Create a release package only after npm run build and npm run tauri:build pass.

Safety

Review file operations before deleting or overwriting files. Review AI responses before inserting generated content into source files.